

Literature status: weak Hilbert spaces need not be UFO

Run `fa_banach_001`; source arXiv:1307.4386

Question and cited source

Castillo–Ferenczi–Moreno’s paper *On Uniformly finitely extensible Banach spaces* records a negative answer to a question from Castillo–Plichko, *Banach spaces in various positions*, J. Funct. Anal. 259 (2010), 2098–2138. The cited question is whether every weak Hilbert space must be UFO.

The answering paper identifies the older article as reference [18].

- [17] J.M.F. Castillo and Y. Moreno, *On the Lindenstrauss-Rosenthal theorem*, Israel J. Math. 140 (2004) 253–270.
- [18] J.M.F. Castillo and A. Plichko, *Banach spaces in various positions*. J. Funct. Anal. 259 (2010) 2098–2138.
- [19] R. Coifman, M. Cwikel, R. Rochberg, Y. Sagher and G. Weiss, *Complex interpolation for*

Figure 1: arXiv:1307.4386, bibliography: reference [18] is the Castillo–Plichko 2010 JFA paper.

Answer recorded in arXiv:1307.4386

The key passage is in Section 4, “Counterexamples.” It recalls the definition of weak Hilbert spaces, considers Tsirelson’s 2-convexified space $\mathcal{T}_2$, and concludes that not all weak Hilbert spaces are UFO.

4. COUNTEREXAMPLES

As we said before, there are not many examples known of either near-Hilbert spaces or spaces with MEP. For a moment, let us focus on B -convex spaces. A good place to look for them is among weak Hilbert spaces. Recall that a Banach space X is said to be weak Hilbert if there exist constants δ, C such that every finite dimensional subspace F contains a subspace G with $\dim G \geq \delta \dim F$ so that $\text{dist}(G, \ell_2^{\dim G}) \leq C$. It is well-known that weak Hilbert spaces are B -convex and near-Hilbert, although it is not known if they must have MEP. Tsirelson’s 2-convexified space $\mathcal{T}_2$ [15] is a weak Hilbert type 2 space, but we do not know whether it is an UFO. Nevertheless, since $\mathcal{T}_2$ is isomorphic to its square we can apply Proposition 3.4 to conclude it is not HUFO. Since subspaces of weak Hilbert spaces are weak Hilbert, we obtain that not all weak Hilbert spaces are UFO. This answers the question left open in [18, p.2131] of whether weak Hilbert spaces must be UFO. Argyros, Beanland and Raikoftsalis [4] have recently constructed a weak Hilbert space X_{abr} with an unconditional basis in which no disjointly supported subspaces are isomorphic (such spaces are called tight by support in [24]) Clearly

Figure 2: arXiv:1307.4386, page 10: the weak-Hilbert counterexample argument and the explicit statement that the question from [18, p.2131] is answered.

Theorem 1 (Literature status). *There exists a weak Hilbert Banach space which is not UFO. Equivalently, the question whether weak Hilbert spaces must be UFO has a negative answer.*

Argument as recorded in the answering paper. The paper uses the following ingredients. First, Tsirelson’s 2-convexified space $\mathcal{T}_2$ is a weak Hilbert type 2 space. Second, $\mathcal{T}_2$ is isomorphic to its square. Third, Proposition 3.4 of the same paper says that any HUFO space isomorphic to its square is isomorphic to a Hilbert space. Since $\mathcal{T}_2$ is not Hilbert, it follows that $\mathcal{T}_2$ is not HUFO.

Now suppose, toward contradiction, that every weak Hilbert space were UFO. Subspaces of weak Hilbert spaces are weak Hilbert, so every closed subspace of $\mathcal{T}_2$ would be UFO. That is exactly to say that $\mathcal{T}_2$ would be HUFO, contradicting the preceding paragraph.

Therefore at least one closed subspace of $\mathcal{T}_2$ is weak Hilbert but not UFO. □

What is not being claimed

The conclusion is existential. It should not be rewritten as “ $\mathcal{T}_2$ is not UFO.” The answering paper explicitly says that it does not know whether $\mathcal{T}_2$ itself is UFO. What it proves is that $\mathcal{T}_2$ is not HUFO, and hence some weak Hilbert subspace of $\mathcal{T}_2$ fails to be UFO.

This packet is also not a new solution by the present run. It records a literature answer already stated in arXiv:1307.4386.

Verification notes and limitations

Local run-index searches on 2026-06-14 found no prior packet for arXiv:1307.4386 or for the weak-Hilbert/UFO question. The packet includes the answering paper and two evidence crops: the answer passage and the bibliography entry identifying [18].

The original Castillo–Plichko 2010 page 2131 was not acquired during this loop. Thus the source-question evidence here is the later authors’ explicit citation and description of the question left open in [18, p.2131].